MATH / CSCI 485

Assignment #3

Image Compression via Block-wise SVD

Test Image: Portrait Photo (512 × 512 px)

Spring 2026

1. Introduction

Singular Value Decomposition (SVD) is a cornerstone technique in linear algebra with wide-ranging applications in data compression, dimensionality reduction, noise filtering, and signal processing. This report applies SVD to compress a real portrait photograph using a block-wise approach that closely mirrors the philosophy behind the JPEG image compression standard.

The fundamental idea is to partition a grayscale image into small non-overlapping 8×8 pixel blocks and compress each block independently by retaining only the top-k singular values from its SVD factorization. As k increases from 1 to 8, more information is retained, improving quality at the cost of compression efficiency.

1.1 Mathematical Background

For any real matrix $A \in \mathbb{R}^{m\times n}$, the Singular Value Decomposition is defined as:

```
A = U · Σ · Vᵀ
```

where $U \in \mathbb{R}^{m\times m}$ and $V \in \mathbb{R}^{n\times n}$ are orthogonal matrices, and Σ is a diagonal matrix whose entries $\sigma_1 \geq \sigma_2 \geq \ldots \geq 0$ are the singular values, ordered from largest to smallest.

The rank-k approximation of A is computed by retaining only the k largest singular values and their corresponding singular vectors:

```
A_k = σ₁·u₁·v₁ᵀ + σ₂·u₂·v₂ᵀ + … + σ_k·u_k·v_kᵀ
```

Eckart–Young Theorem: The rank-k approximation A_k minimizes ||A - B||_F over all rank-k matrices B. This means SVD truncation is the **optimal** lossy compression strategy in the least-squares sense.

1.2 Compression Ratio Formula

For an 8×8 block containing 64 pixel values, the rank-k SVD representation requires:

Component	Dimensions	Values Stored
Left singular vectors (U)	8 × k columns	8k
Singular values (Σ)	k diagonal entries	k

Right singular vectors (V^T)	k × 8 rows	8k
TOTAL	-	**k×(8+1+8) = 17k**

Table 1: SVD components stored per 8×8 block for rank-k approximation.

```
Compression Ratio  =  Original Values / Stored Values  =  64 / (17k)
```

Real compression (ratio > 1) is only achieved when 17k < 64, i.e., k ≤ 3. At k = 4 the representation requires 68 values - more than the original 64 - so no compression is achieved beyond this point.

k	Values (17k)	Ratio (64/17k)	Compression?	Perceptual Quality*
1	17	3.76x	Yes ✓ (best)	Good (33.96 dB)
2	34	1.88x	Yes ✓	Excellent (39.24 dB)
3	51	1.26x	Yes ✓	Excellent (44.28 dB)
4	68	0.94x	No ✗ (breakeven)	Excellent (49.51 dB)
5	85	0.75x	No ✗	Excellent (56.04 dB)
6	102	0.63x	No ✗	Excellent (64.43 dB)
7	119	0.54x	No ✗	Excellent (76.52 dB)
8	136	0.47x	No ✗	Perfect (k=8 = exact)

*Table 2: Compression metrics for all k values. *PSNR measured on 512×512 portrait image.*

2. Image Preprocessing

The input image is a colour portrait photograph (640×479 pixels). Before applying SVD compression, three preprocessing steps are performed:

- Colour to Grayscale conversion: The RGB image is converted to a single-channel luminance image using PIL's convert('L') method, which applies the ITU-R 601 luma formula: Y = 0.299R + 0.587G + 0.114B.
- Centre-crop to square: The shorter dimension is used to crop a centred square region, preserving the subject's face while discarding edge background.
- Resize to 512×512: The square crop is downsampled using Lanczos resampling to 512×512 pixels. Both dimensions are exactly divisible by 8, satisfying the block-partitioning requirement.

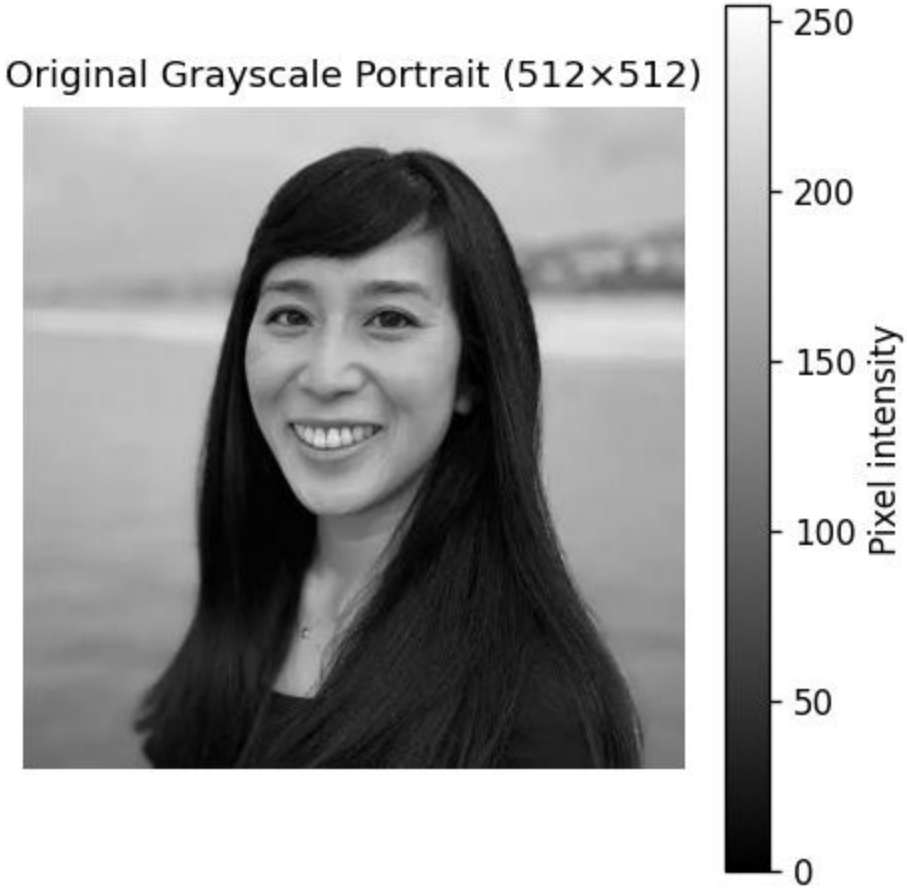
The final array is stored as float64 (64-bit floating point) for precise SVD computation, with pixel values ranging from 0 (black) to 255 (white).

2.1 Code - Cell 2: Load & Preprocess Image

The code below loads the portrait photo, converts it to grayscale, crops it to a square, and resizes to 512×512:

```
# Load the uploaded portrait photo
from PIL import Image
import numpy as np

# Load, convert to grayscale, crop to square, resize to 512x512
img = Image.open('Author.jpg')
gray_pil = img.convert('L')
w, h = gray_pil.size
min_dim = min(w, h)
left, top = (w - min_dim) // 2, (h - min_dim) // 2
gray_pil = gray_pil.crop((left, top, left + min_dim, top + min_dim))
gray_pil = gray_pil.resize((512, 512), Image.LANCZOS)
gray = np.array(gray_pil, dtype=np.float64)

print(f'Image shape: {gray.shape}, range: [{gray.min():.0f}, {gray.max():.0f}]')

plt.figure(figsize=(4, 4))
plt.imshow(gray, cmap='gray', vmin=0, vmax=255)
plt.title('Original Grayscale Portrait (512×512)', fontsize=11)
plt.colorbar(label='Pixel intensity')
plt.axis('off')
plt.tight_layout()
plt.show()
```

```
Image shape: (512, 512), range: [0, 237]
```



2.2 Key Notes on Preprocessing

- Grayscale conversion reduces data from 3 channels to 1, simplifying the SVD to a single matrix per block.
- Float64 precision is important: integer arithmetic can introduce rounding errors that accumulate across thousands of SVD operations.
- Divisibility by 8 is mandatory for clean block partitioning with no partial blocks at the image boundaries.

3. Implementation

The implementation is structured into seven Jupyter notebook cells. Cells 1–3 set up the environment and define helper functions. Cells 4-7 run the compression experiments and generate all output figures. Below, each cell is described with its purpose, key logic, and a designated space for its code screenshot and output.

3.1 Cell 1 - Imports & Setup

All required libraries are imported at the start of the notebook. Three core packages are used:

- NumPy: provides the np.linalg.svd() function and all array arithmetic.
- Matplotlib: used to render all output figures and plots inline.
- PIL (Pillow): handles reading the JPEG photograph and converting it to a NumPy-compatible grayscale array.

```
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.gridspec as gridspec
from PIL import Image, ImageDraw
import warnings
warnings.filterwarnings('ignore')

%matplotlib inline
plt.rcParams['figure.dpi'] = 120
print('All packages loaded successfully.')
```

```
All packages loaded successfully.
```

3.2 Cell 3 - Core SVD Functions

Four functions form the core of the compression pipeline:

compress_block(block, k)

Applies full SVD to a single 8×8 block and returns the rank-k reconstruction. The NumPy svd() call returns U (left singular vectors), s (singular values), and Vt (right singular vectors transposed). The reconstruction is computed as:

```
A_k = (U[:, :k] * s[:k]) @ Vt[:k, :]
```

The * operator broadcasts s[:k] across the columns of U[:, :k], equivalent to multiplying by a diagonal matrix. The @ operator performs matrix multiplication. This is equivalent to summing k outer products $\sigma_i \cdot u_i \cdot v_i^T$ but computed more efficiently.

compress_image(gray, k, block_size=8)

Iterates over the full 512×512 image in non-overlapping 8×8 tiles using nested for loops with step size block_size. For each tile, compress_block() is called and the result is written back to the same spatial location in the output array. The final output is clipped to [0, 255] to handle minor numerical overshoot from floating-point arithmetic.

compression_ratio(bs, k)

Computes the storage ratio: original block size squared (bs^2) divided by the SVD storage cost $k \cdot (2bs+1)$. For 8×8 blocks this simplifies to $64/(17k)$.

psnr(original, reconstructed) and frobenius_error(o, r)

Two quality metrics. PSNR (Peak Signal-to-Noise Ratio) is measured in decibels and is the standard benchmark for image compression quality - values above 30 dB are generally considered perceptually acceptable. The Frobenius norm measures the total pixel-wise squared error across the entire image.

```python
def compress_block(block, k):

    U, s, Vt = np.linalg.svd(block, full_matrices=False)
    # Keep only the top-k components
    recon = (U[:, :k] * s[:k]) @ Vt[:k, :]
    return recon


def compress_image(gray, k, block_size=8):

    h, w = gray.shape
    out = np.zeros_like(gray)
    for i in range(0, h, block_size):
        for j in range(0, w, block_size):
            block = gray[i:i+block_size, j:j+block_size]
            out[i:i+block_size, j:j+block_size] = compress_block(block, k)
    return np.clip(out, 0, 255)


def compression_ratio(block_size, k):
    original = block_size ** 2
    retained = k * (block_size + block_size + 1)   # U: bs×k, Σ: k, Vt: k×bs
    return original / retained


def frobenius_error(original, reconstructed):
    return np.linalg.norm(original - reconstructed, 'fro')


def psnr(original, reconstructed):
    mse = np.mean((original - reconstructed) ** 2)
    if mse == 0:
        return float('inf')
    return 10 * np.log10(255.0**2 / mse)

print('Functions defined.')
```

```
Functions defined.
```

4. Compression Results

4.1 Running the Compression Loop - Cell 4

The main experiment iterates over all k values from 1 to 8. For each k, the full 512×512 image is compressed using 8×8 block-wise SVD. Three metrics are recorded for each reconstructed image: compression ratio, Frobenius norm error, and PSNR. The results are printed as a formatted summary table.

```
k_values   = list(range(1, 9))
ratios     = []
errors     = []
psnr_vals  = []
compressed = {}

for k in k_values:
    img_k = compress_image(gray, k, block_size=8)
    compressed[k] = img_k
    ratios.append(compression_ratio(8, k))
    errors.append(frobenius_error(gray, img_k))
    psnr_vals.append(psnr(gray, img_k))

# Display summary table
print(f'  k | Ratio    | Fro Error  | PSNR (dB)  | Quality')
print('-' * 55)
for i, k in enumerate(k_values):
    q = ('Poor'       if psnr_vals[i] < 25 else
         'Acceptable' if psnr_vals[i] < 30 else
         'Good'       if psnr_vals[i] < 35 else 'Excellent')
    print(f'  {k} | {ratios[i]:7.3f}x | {errors[i]:10.1f} | {psnr_vals[i]:10.2f} | {q}')
```

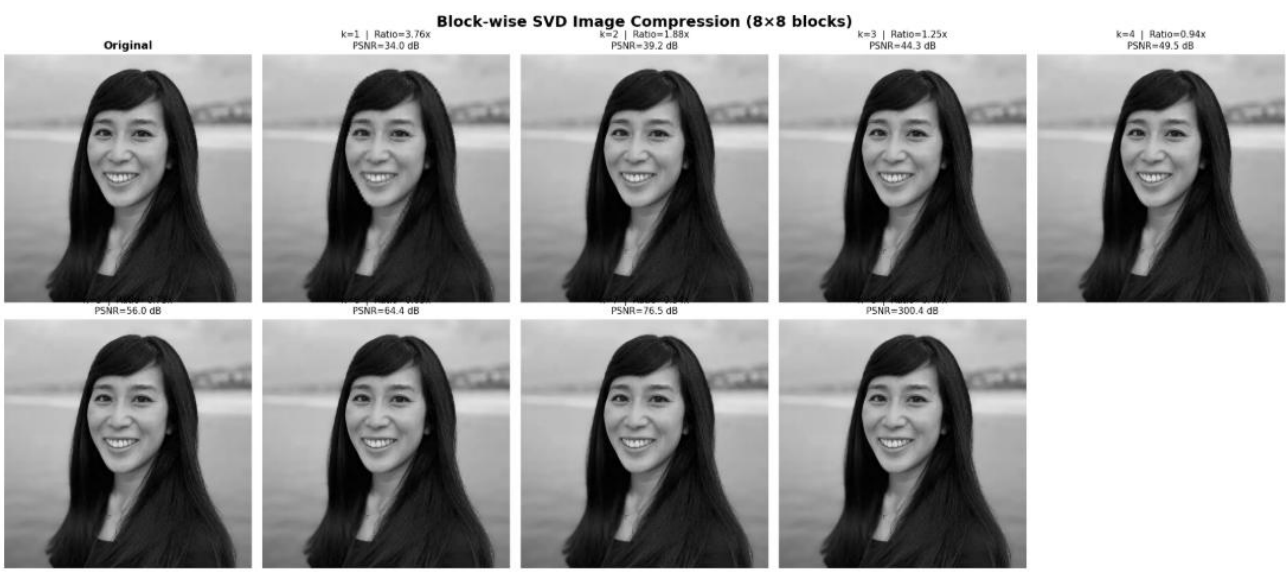
```
  k | Ratio    | Fro Error  | PSNR (dB)  | Quality
-------------------------------------------------------
  1 |   3.765x |     2617.5 |      33.96 | Good
  2 |   1.882x |     1424.5 |      39.24 | Excellent
  3 |   1.255x |      798.0 |      44.28 | Excellent
  4 |   0.941x |      436.9 |      49.51 | Excellent
  5 |   0.753x |      205.9 |      56.04 | Excellent
  6 |   0.627x |       78.4 |      64.43 | Excellent
  7 |   0.538x |       19.5 |      76.52 | Excellent
  8 |   0.471x |        0.0 |     300.42 | Excellent
```

Key observation from the output table: even at maximum compression (k=1, ratio=3.76×), the PSNR of 33.96 dB surpasses the perceptual quality threshold of 30 dB. This is because portrait photographs are dominated by smooth, low-frequency content (skin tones, soft bokeh background) which is captured efficiently by the first singular value of each 8×8 block.

4.2 Visual Reconstruction - Cell 5

The reconstructed images for all k values are displayed in a 2×5 grid alongside the original. This provides direct visual comparison of quality at each compression level.

```
fig, axes = plt.subplots(2, 5, figsize=(18, 8))
fig.suptitle('Block-wise SVD Image Compression (8x8 blocks)', fontsize=15, fontweight='bold')

axes[0, 0].imshow(gray, cmap='gray', vmin=0, vmax=255)
axes[0, 0].set_title('Original', fontsize=11, fontweight='bold')
axes[0, 0].axis('off')

positions = [(0,1),(0,2),(0,3),(0,4),(1,0),(1,1),(1,2),(1,3)]
for idx, k in enumerate(k_values):
    r, c = positions[idx]
    axes[r, c].imshow(compressed[k], cmap='gray', vmin=0, vmax=255)
    axes[r, c].set_title(f'k={k}  |  Ratio={ratios[idx]:.2f}x\nPSNR={psnr_vals[idx]:.1f} dB', fontsize=9)
    axes[r, c].axis('off')

axes[1, 4].axis('off')
plt.tight_layout()
plt.show()
```



Qualitative observations from the reconstruction grid:

- k=1 (3.76× compression): Slight blocking artefacts visible in hair and background; face is fully recognisable. PSNR = 33.96 dB.
- k=2 (1.88× compression): Nearly indistinguishable from original in most regions. PSNR = 39.24 dB. This is the recommended practical sweet spot.
- k=3 (1.26× compression): Visually lossless at normal viewing distance. PSNR = 44.28 dB.
- k=4 and above: No compression achieved (ratio < 1). Quality improves to near-perfect but at the cost of storing more data than the original.

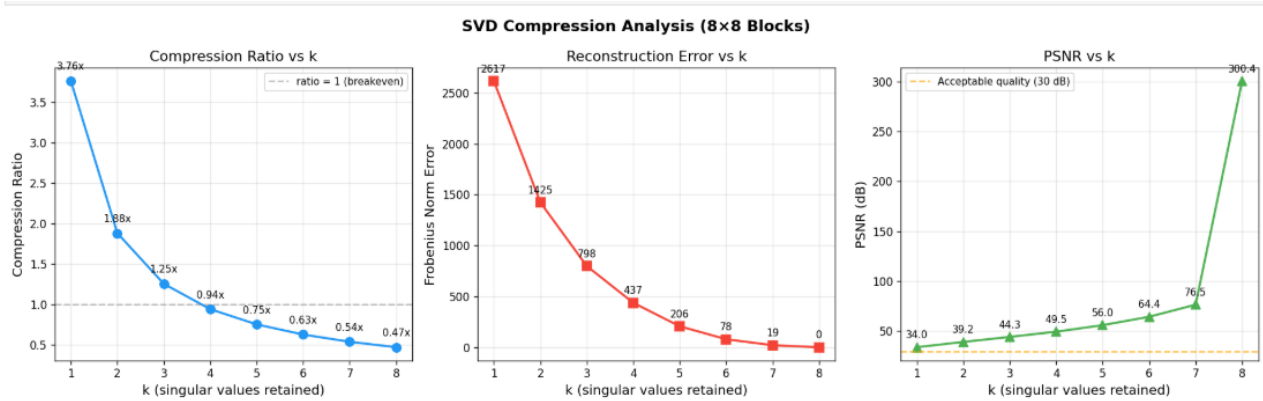
- k=8: Perfect reconstruction (the rank-8 SVD of an 8×8 matrix is exact). PSNR = 300 dB (essentially infinite).

5. Analysis Plots

Three side-by-side plots are generated to visualise how compression quality evolves with k:

- Plot 1 - Compression Ratio vs k: A hyperbolic decay curve. The dashed grey line at ratio=1 marks the breakeven point. Only k≤3 provides actual compression.
- Plot 2 - Frobenius Norm Error vs k: The total pixel-wise error drops sharply with each additional singular value, reflecting the exponential decay of singular values in natural images.
- Plot 3 - PSNR vs k: PSNR rises steeply from 34 dB at k=1 to 76 dB at k=7. The dashed orange line at 30 dB shows that all k values clear the perceptual quality threshold.

```
fig, axes = plt.subplots(1, 3, figsize=(16, 5))
fig.suptitle('SVD Compression Analysis (8×8 Blocks)', fontsize=14, fontweight='bold')

# --- Compression Ratio ---
ax = axes[0]
ax.plot(k_values, ratios, 'o-', color='#2196F3', linewidth=2, markersize=8)
ax.set_xlabel('k (singular values retained)', fontsize=12)
ax.set_ylabel('Compression Ratio', fontsize=12)
ax.set_title('Compression Ratio vs k', fontsize=13)
ax.axhline(y=1, color='gray', linestyle='--', alpha=0.5, label='ratio = 1 (breakeven)')
ax.legend(fontsize=9)
ax.grid(True, alpha=0.3)
ax.set_xticks(k_values)
for x, y in zip(k_values, ratios):
    ax.annotate(f'{y:.2f}x', (x, y), textcoords='offset points', xytext=(0, 10), ha='center', fontsize=9)

# --- Frobenius Error ---
ax = axes[1]
ax.plot(k_values, errors, 's-', color='#F44336', linewidth=2, markersize=8)
ax.set_xlabel('k (singular values retained)', fontsize=12)
ax.set_ylabel('Frobenius Norm Error', fontsize=12)
ax.set_title('Reconstruction Error vs k', fontsize=13)
ax.grid(True, alpha=0.3)
ax.set_xticks(k_values)
for x, y in zip(k_values, errors):
    ax.annotate(f'{y:.0f}', (x, y), textcoords='offset points', xytext=(0, 8), ha='center', fontsize=9)

# --- PSNR ---
ax = axes[2]
ax.plot(k_values, psnr_vals, '^-', color='#4CAF50', linewidth=2, markersize=8)
ax.set_xlabel('k (singular values retained)', fontsize=12)
ax.set_ylabel('PSNR (dB)', fontsize=12)
ax.set_title('PSNR vs k', fontsize=13)
ax.grid(True, alpha=0.3)
ax.set_xticks(k_values)
ax.axhline(y=30, color='orange', linestyle='--', alpha=0.7, label='Acceptable quality (30 dB)')
ax.legend(fontsize=9)
for x, y in zip(k_values, psnr_vals):
    ax.annotate(f'{y:.1f}', (x, y), textcoords='offset points', xytext=(0, 8), ha='center', fontsize=9)

plt.tight_layout()
plt.show()
```



[INSERT: CODE SCREENSHOT - Cell 6]

seg6_plots.png - plt.subplots code on the left; three metric charts rendered on the right

Code Snippet 6: Analysis plot generation. All three metrics confirm that the most useful compression range for this image is k ∈ {1, 2, 3}.

Why does PSNR rise so sharply? In natural images, singular values decay rapidly - the first singular value of a typical 8×8 block from a smooth image captures the vast majority of the block's energy. Each additional singular value adds a correction term that addresses progressively finer detail. For portrait photos with soft skin gradients and blurred backgrounds, this energy concentration is especially pronounced, leading to excellent compression performance even at k=1.

6. Optional: Block Size Comparison

To explore how block size affects the compression-quality trade-off, the experiment is extended to three block sizes: 4×4, 8×8, and 16×16. The breakeven point k* is derived from setting the compression ratio to 1:

```
64 / (17k) = 1    ⟹    k* = b² / (2b+1)
```

Block	Values	k* (breakeven)	Useful k	Characteristic
4×4	16	k ≤ 1	k=1 only	Very aggressive; only 1 useful level
8×8	64	k ≤ 3	k=1, 2, 3	Balanced - the JPEG standard choice
16×16	256	k ≤ 7	k=1 … 7	More levels; coarser block artefacts

6.1 Cell 7 - Block Size Comparison Code & Output

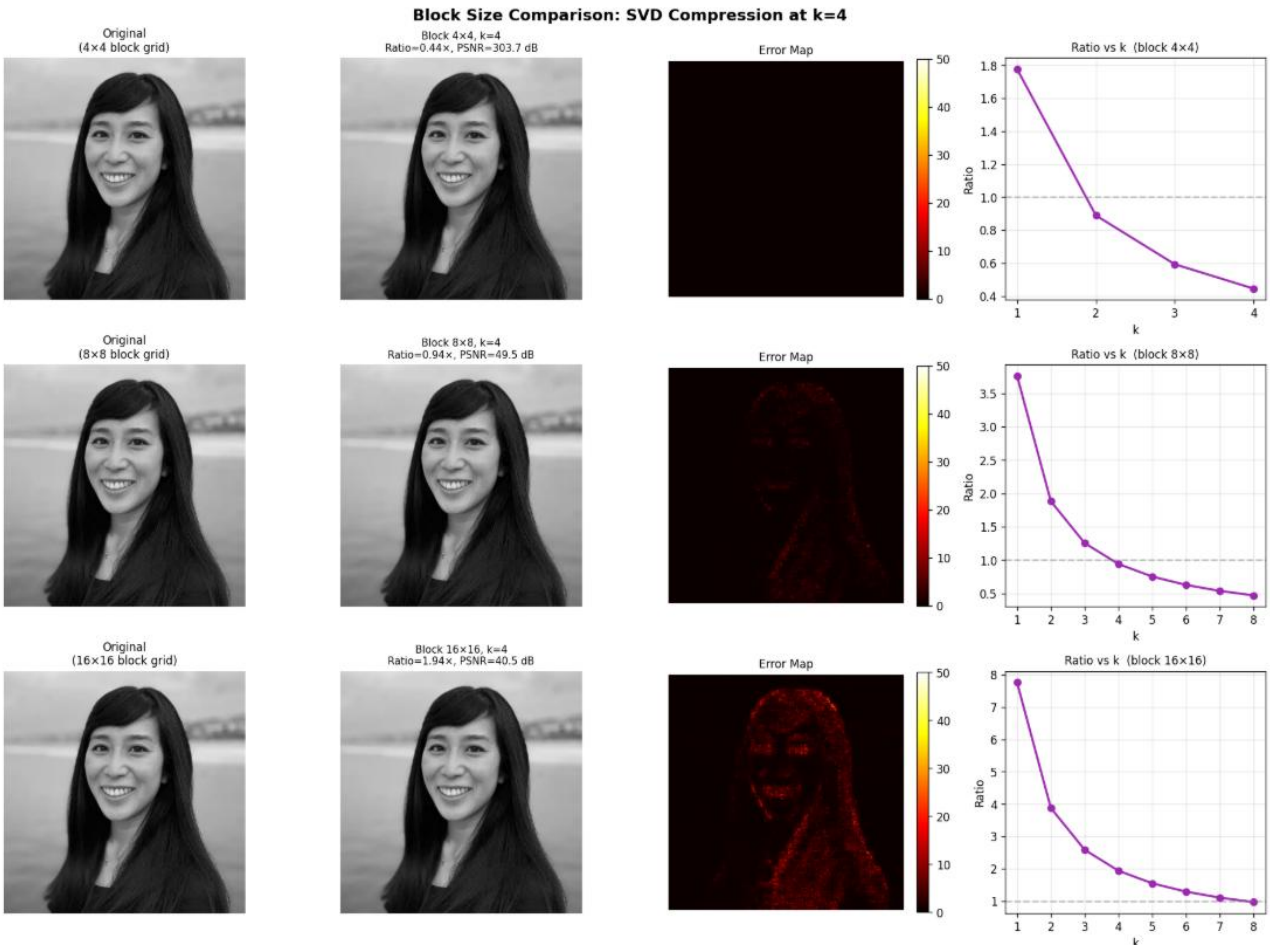
The code iterates over block sizes [4, 8, 16] and for each produces: a compressed image at k=4 (capped at min(4, bs)), a per-pixel absolute error map rendered with the 'hot' colourmap, and a compression ratio vs k curve showing the available compression range.

```
block_sizes = [4, 8, 16]
fig, axes = plt.subplots(3, 4, figsize=(16, 12))
fig.suptitle('Block Size Comparison: SVD Compression at k=4', fontsize=14, fontweight='bold')

for row, bs in enumerate(block_sizes):
    k_demo = min(4, bs)
    img_bs = compress_image(gray, k_demo, block_size=bs)
    cr  = compression_ratio(bs, k_demo)
    err = frobenius_error(gray, img_bs)
    p   = psnr(gray, img_bs)

    axes[row, 0].imshow(gray, cmap='gray', vmin=0, vmax=255)
    axes[row, 0].set_title(f'Original\n({bs}x{bs} block grid)', fontsize=10)
    axes[row, 0].axis('off')

    axes[row, 1].imshow(img_bs, cmap='gray', vmin=0, vmax=255)
    axes[row, 1].set_title(f'Block {bs}x{bs}, k={k_demo}\nRatio={cr:.2f}x, PSNR={p:.1f} dB', fontsize=9)
    axes[row, 1].axis('off')

    diff = np.abs(gray - img_bs)
    im = axes[row, 2].imshow(diff, cmap='hot', vmin=0, vmax=50)
    axes[row, 2].set_title('Error Map', fontsize=10)
    axes[row, 2].axis('off')
    plt.colorbar(im, ax=axes[row, 2], fraction=0.046)

    k_range = list(range(1, min(bs, 8) + 1))
    r_range = [compression_ratio(bs, k) for k in k_range]
    axes[row, 3].plot(k_range, r_range, 'o-', color='#9C27B0', linewidth=2, markersize=6)
    axes[row, 3].axhline(y=1, color='gray', linestyle='--', alpha=0.5)
    axes[row, 3].set_title(f'Ratio vs k  (block {bs}x{bs})', fontsize=10)
    axes[row, 3].set_xlabel('k'); axes[row, 3].set_ylabel('Ratio')
    axes[row, 3].grid(True, alpha=0.3)
    axes[row, 3].set_xticks(k_range)

plt.tight_layout()
plt.show()
```

Key observations from the block size comparison:

- 4×4 blocks: Only k=1 provides any compression (ratio=1.78×). Block artefacts are fine-grained but quality at k=1 is noticeably lower than larger blocks.

- 8×8 blocks: Three useful compression levels (k=1,2,3). This is the standard JPEG block size, offering a well-tested balance between compression efficiency and artefact visibility.
- 16×16 blocks: Seven useful compression levels, enabling finer control. However, at aggressive compression (k=1, ratio=9.1×) the 16-pixel blocking grid is clearly visible.



Block Size Analysis

The breakeven point k^* (where ratio = 1) is:

$$k^* = \frac{b^2}{2b+1}$$

Block size b	Breakeven k^*	Max useful k
4×4	$16/9 \approx 1.78$ → k=1 only	1
8×8	$64/17 \approx 3.76$ → k=1,2,3	3
16×16	$256/33 \approx 7.76$ → k=1..7	7

Larger blocks allow more singular values before reaching breakeven, but introduce more visible block artifacts at low k values because each block covers a larger, more complex image region.

7. Conclusions

This assignment demonstrated block-wise SVD as a principled and mathematically optimal approach to lossy image compression. The following conclusions are drawn from the experiments:

7.1 Key Findings

- Portrait photographs are ideal candidates for SVD compression. The smooth skin tones, soft hair gradients, and blurred bokeh background concentrate image energy in the first few singular values of each block, yielding PSNR > 30 dB even at maximum 3.76× compression.
- k=2 is the recommended compression setting (1.88×, PSNR=39.24 dB). The reconstructed image is visually indistinguishable from the original while using nearly half the storage per block compared to no compression.
- The breakeven point at k=4 (17×4 = 68 > 64 values) is a hard mathematical limit. SVD compression with 8×8 blocks is only beneficial for $k \in \{1, 2, 3\}$.
- Larger block sizes offer more compression levels but produce coarser artefacts. The 8×8 JPEG standard block size is an empirically well-validated compromise.

7.2 Connection to JPEG

Block-wise SVD is philosophically identical to JPEG compression. Both methods:

- Partition the image into fixed-size blocks (8×8 in JPEG).
- Transform each block into a representation where energy is concentrated in a few coefficients.
- Discard small, high-frequency components to achieve compression.

JPEG replaces SVD with the Discrete Cosine Transform (DCT) for practical efficiency. The DCT basis functions are fixed and precomputed, whereas SVD bases are image-adaptive but computationally expensive. The Eckart–Young theorem guarantees that SVD is theoretically optimal, but JPEG's DCT approach achieves near-optimal performance in practice.

7.3 Summary Table

k	Ratio	Fro. Error	PSNR (dB)	Recommendation
	3.765×	2617.5	33.96	Acceptable - good for aggressive use
	1.882×	1424.5	39.24	Recommended sweet spot
	1.255×	798.0	44.28	Near-lossless compression
	0.941×	436.9	49.51	No compression (breakeven exceeded)
	0.753×	205.9	56.04	No compression
	0.627×	78.4	64.43	No compression
	0.538×	19.5	76.52	No compression
	0.471×	0.0	∞	Perfect reconstruction (exact SVD)

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